

Shihao Cao

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It is only with the heart that one can see rightly – Antoine de Saint-Exupéry

EDUCATION

CORNELL UNIV.

GPA: 3.95 / 4.0

Computer Science

Dec 2022 | Ithaca, NY

THOMAS JEFFERSON HS FOR SCI/TECH

GPA: 4.56 / 4.0

Jun 2019 | Alexandria, VA

LINKS

Website:// shihaoceo.com

LinkedIn:// [shihaoceo](https://www.linkedin.com/in/shihaoceo)

Github:// [shihaoceo](https://github.com/shihaoceo)

Twitter:// [shihao_ceo](https://twitter.com/shihao_ceo)

SKILLS

PROGRAMMING

skilled:

Python • SQL • C++

HOOTL/HITL Testing

RC Aircraft (*Design, Build, Pilot*)

Embedded SW • Sensors

I2C • Serial • Arduino

Simulation • Flight Software

OpenCV (*Python and C++*)

experienced:

C • Anyio • NumPy

Prometheus • Networking

COURSEWORK

OS + Prac. (CS 4410)

Compilers + Prac. (CS 4120)

Computer Arch. (ECE 4750)

Adv. Systems (CS 6410)

Algorithms (CS 4820)

Adv. ML Systems (CS 6787)

Intro ML (CS 4780)

Numerical Comp. (CS 4220)

Computer Sys. (CS 3410)

Computer Vision (CS 4670)

Functional Progr. (CS 3110)

Discrete Math (CS 2800)

OOP & DataS. (CS 2110)

Probability (MATH 4710)

HIGH SCHOOL:

ML • CV • AI • Robotics

Linear • DiffEq • MultiVar

Quantum Mechanics

HOBBIES

Filmmaking • Longboarding

RC Aircraft • 24h of Lemons

EXPERIENCE

SPACEX | STARLINK SOFTWARE ENGINEER (DIRECT-TO-CELL CORE NETWORK)

Jan 2024 - Now | Sunnyvale, CA

- Owned and built **Wireless Emergency Alerts from space** for DTC Starlink. We can serve AMBER, tsunami/earthquake, etc. alerts anywhere over USA/Canada/Japan in under 1s.
- Built the entire HOOTL satellite/phone integration suite with async Python. Runs in 5 mins.
- Scaled and bugfixed C++ LTE software to serve from 0 to thousands of users with T-Mobile

ORCHARD ROBOTICS | Co-FOUNDER AND CTO

Nov 2022 - Dec 2023 | Ithaca, NY

- Led 2 engineers and 4 interns, developing a Nvidia Jetson-based **Orchard Vision System**
- Built 15+ systems to scan thousands of acres, yielding within 5% accurate fruit counts & sizes
- Architected image stitching pipeline to tag each fruit to within 1m of its tree
- Deployed weekly to 7 Orchards across the US to develop and test with farm managers on site
- Scaled from first demo to raising seed round backed by General Catalyst

JUMP TRADING | SOFTWARE ENGINEERING INTERN

Jun 2022 - Aug 2022 | Chicago, IL

- Built custom cache aware low latency (ns range) high rate (10Gbps+) stats framework in C++
- Architected data flow thru 100+ apps, 10+ hosts, shared mem, databases, and frontends
- Deployed stats framework into two production clusters, 10x-ing existing stats data flow

SPACEX | STARSHIP FLIGHT SOFTWARE INTERN

Jun 2021 - Aug 2021 | Hawthorne, CA

- Owned, and supported inter-process data sharing for flight software (200+ devs)
- Architected flight software abort relay verification system, catching 4 critical bugs
- Optimized generation speed by 100x through data caching and multi-threaded workflows

SPACEX | VEHICLE ENGINEERING INTERN

May 2020 - Aug 2020 | Hawthorne, CA

- Responsible for production landscape and engineering of two valves on Falcon 9 and Merlin
- Iterated design and process to eliminate 90% of rebuilds, doubling production rate

RESEARCH AND TEAMS

SPACE SYSTEMS DESIGN STUDIO - PAN TEAM | Co-LEAD

Sept 2019 - May 2022 | Ithaca, NY

- Spearheaded software development for the Pathfinder for Autonomous Navigation (PAN) project, two 3U Cube Satellites which will autonomously rendezvous and dock in LEO
- Developed C++ drivers and flight software for attitude control system and GPS
- Devised HITL/HOOTL tests to validate drivers, state-machines, and hardware

TJ UNMANNED AERIAL VEHICLE TEAM | PRESIDENT + FOUNDER

Sept 2017 - Jun 2019 | Alexandria, VA

- In one year, lead team of 20 to design and build the **2019 fixed-wing UAV** for SUAS
- Spearheaded flight software, power system, CONOPS, and mech. design/manufacturing

PERSONAL PROJECTS

1997 TOYOTA AVALON - 24 HRS OF LEMONS - TEAM CAPTAIN | MAY 2025

Lead a team of 12 to fix and race a 1997 Avalon for a **25-hr endurance race**, won Judge's Choice

7FT 7IN TOWER OF CARDS | MAY 2022

I wanted to build something I remembered being amazed by as a child – a **massive tower of cards**

ELECTRIC STARSHIP HOPPER | MAY 2020

With a partner, we designed and built an **electric VTOL drone** with single point vectored thrust.

I designed the vehicle, wrote the C++ flight software, sensor/action pipeline, and telemetry flow